

Notes: Waldinger (JPE, 2010)
**Quality Matters: The Expulsion of Professors and the Consequences for PhD
Student Outcomes in Nazi Germany**

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	PhD student data	4
3.2	Dismissed professor data	4
3.3	Professor roster and department quality	5
3.4	Treatment variables	5
4	Models and empirical specifications	5
4.1	Baseline outcome equation	5
4.2	First-stage equations	6
4.3	Reduced-form model	6
4.4	Instrumental variables interpretation	7
5	Identification and empirical design	7
5.1	What creates identification	7
5.2	Main exclusion concern	7
5.3	Threats and responses	8
5.4	What the design identifies	8
6	Tables and figures	8
6.1	Figure 1 and Table 1: Motivation and timing of the shock	8
6.2	Table 2 and Table 3: Cross-department variation and quality of dismissed faculty . .	9
6.3	Table 4 and Figure 2: Student data and first-stage visual evidence	10
6.4	Figure 3 and Table 5: Reduced-form outcomes and first stages	11
6.5	Table 6 and Table 7, part 1: Main IV estimates and robustness	12
6.6	Table 7, part 2 and Table 8: Robustness continued and heterogeneity	13
6.7	Appendix Table A1 and Appendix Table A2: Measurement details	14

6.8	Appendix Table A3 and Appendix Figure A1: Reduced form, placebo, and source material	15
6.9	Appendix Figure A2: Jewish PhD students	16
7	Short summary	17
8	Conclusion	18
8.1	Core conclusion	18
8.2	Economic intuition	18
8.3	Incremental contribution revisited	18
8.4	Implications for PhD training and science policy	18
8.5	Limitations	18
8.6	Takeaway	19

1 Big picture

- **Question.** Does the quality of faculty in a PhD department causally affect PhD student outcomes? The paper studies whether better faculty improve students' short-run research output and long-run academic careers.
- **Core empirical challenge.** High-quality students select into high-quality departments, and high-quality departments may also have better labs, networks, funding, and reputations. A simple correlation between department quality and student success is therefore not enough.
- **Natural experiment.** The paper uses the expulsion of Jewish and politically unreliable mathematics professors in Nazi Germany as an exogenous shock to department quality. The dismissals occurred sharply after 1933 and hit departments very unevenly.
- **Setting.** The analysis focuses on mathematics PhD students in Germany who received their degrees between 1923 and 1938. This period is useful because the students' entire academic careers can be followed.
- **Treatment.** Some departments, especially high-quality ones such as Göttingen and Berlin, lost a large fraction of their faculty. Other departments were largely unaffected.
- **Faculty-quality measure.** Department quality is measured using the average citation-weighted publication output of mathematics professors before the dismissals.
- **Main outcomes.** The paper studies whether a PhD student publishes the dissertation in a top journal, later becomes a full professor, receives lifetime citations, and has any positive lifetime citations.
- **Main result.** Faculty quality has large effects. A one-standard-deviation increase in faculty quality raises the probability of publishing the dissertation in a top journal by about 13 percentage points, the probability of becoming a full professor by about 10 percentage points, lifetime citations by about 6.3, and the probability of having positive lifetime citations by about 16 percentage points.
- **Quantity versus quality.** The student/faculty ratio does not have a robust effect on the measured PhD student outcomes. The results point to faculty quality rather than just faculty quantity.
- **Economic takeaway.** PhD training is not merely a signal or a sorting device. The quality of the intellectual environment appears to create human capital that matters for research careers.

2 Incremental contribution of the paper

- **From correlation to causal evidence.** Earlier work on PhD outcomes often documents that students from better programs do better. This paper uses a historical shock to department quality to move closer to causal interpretation.
- **A clean quality shock.** The expulsion of professors provides department-level variation in faculty quality that was not chosen by students or departments in response to expected PhD outcomes.
- **Direct measurement of faculty quality.** The paper constructs a professor-by-year and department-by-year quality measure from citation-weighted publications in top journals.

- **Long-run outcomes.** Because the historical cohorts had completed their careers, the paper can study lifetime citations and full-professor status, not only early publications.
- **Quality versus congestion.** The paper separates the effect of average faculty quality from the effect of the student/faculty ratio. This distinguishes intellectual quality from simple resource congestion.
- **New historical microdata.** The data combine the universe of German mathematics PhD students, a roster of mathematics professors, dismissal records, and publication/citation data.
- **Identification diagnostics.** The paper performs robustness checks for Jewish students, disruption during 1933–1934, advisor-loss channels, department-specific trends, and placebo dismissals before 1933.
- **Policy contribution.** The results imply that concentrating PhD training in high-quality departments can generate meaningful returns for student careers and scientific output.

3 Data and measurement

3.1 PhD student data

- The student data contain the universe of mathematics PhD students who received their PhD from a German university between 1923 and 1938.
- The paper focuses on completed PhDs before World War II, which avoids the major disruption after 1939 and allows long-run outcomes to be observed.
- The data include students’ university histories, dissertation information, and later careers.
- The baseline analysis contains 690 PhD students.
- The main short-run outcome is an indicator for whether the dissertation was published in a top journal.
- The main long-run outcomes are whether the student became a full professor, the number of lifetime citations in mathematics journals, and an indicator for any positive lifetime citations.

Interpretation: The student data are unusually strong because they combine complete historical coverage with career outcomes. The limitation is that the analysis observes completed PhDs, not the full population of entrants who may have dropped out or changed paths.

3.2 Dismissed professor data

- The main dismissal source is the *List of Displaced German Scholars*, supplemented with biographical and historical sources.
- The Nazi civil-service law of 1933 led to the dismissal of Jewish and politically unreliable professors.
- In mathematics, 41 professors were dismissed in 1933–1934, about 18.3% of the mathematics professoriate at the beginning of 1933.
- Dismissals were not evenly spread across departments. Göttingen lost 10 of 17 professors, while many departments lost none.

- The dismissed professors were, on average, younger and much more productive than the stayers.

Interpretation: The dismissal shock changed both the number and the average quality of faculty. This is why the paper needs instruments for both faculty quality and the student/faculty ratio.

3.3 Professor roster and department quality

- The paper constructs a yearly roster of mathematics professors at German universities from the semiofficial University Calendar.
- Department quality is the average of individual professor productivity measures among professors currently present in that department-year.
- Individual professor productivity is measured using citation-weighted publications in top journals between 1925 and 1932.
- Publications are drawn from mathematics journals and important related journals in general science, physics, and chemistry.
- Because the productivity measure is fixed using pre-1933 output, it is not mechanically affected by the dismissals themselves.

Interpretation: The department-quality variable captures the intellectual strength of the local research environment that students face. It is imperfect, but the historical citation measure gives a transparent and comparable ranking of departments.

3.4 Treatment variables

- The first instrument is the *dismissal-induced reduction in faculty quality*. It measures how much average department quality fell when above-average-quality professors were dismissed.
- The second instrument is the *dismissal-induced increase in the student/faculty ratio*. It measures the mechanical increase in congestion caused by losing professors.
- For students finishing after 1933, the relevant department is assigned based on the department the student attended at the beginning of 1933. This avoids endogenous post-shock switching.

Interpretation: These instruments translate the historical dismissals into two empirical shocks: one to average faculty quality and one to department capacity.

4 Models and empirical specifications

4.1 Baseline outcome equation

The basic student-level model is

$$\begin{aligned} Outcome_{idt} = & \beta_0 + \beta_1 Q_{dt} + \beta_2 SFR_{dt} + \beta_3 Female_{idt} + \beta_4 Foreigner_{idt} \\ & + \lambda_t + \mu_d + \varepsilon_{idt}, \end{aligned}$$

where Q_{dt} denotes average faculty quality and SFR_{dt} denotes the student/faculty ratio.

- $Outcome_{idt}$ is one of the PhD student outcomes for student i in department d and PhD cohort t .
- $AvgFacultyQuality_{dt}$ is the department's average citation-weighted faculty quality.
- $StudentFacultyRatio_{dt}$ measures the size of the potential student cohort relative to the number of professors.
- Cohort fixed effects absorb shocks common to all students graduating in a given year.
- Department fixed effects absorb time-invariant differences across departments.

Interpretation: OLS estimates of β_2 are likely biased because students sort into departments and because faculty quality is measured with error. The IV design aims to isolate changes in quality caused by dismissals.

4.2 First-stage equations

The first-stage equation for faculty quality is

$$Q_{dt} = \gamma_0 + \gamma_1 DQ_{dt} + \gamma_2 DSFR_{dt} + \Gamma X_{idt} + \lambda_t + \mu_d + u_{idt},$$

where DQ_{dt} is the dismissal-induced fall in faculty quality and $DSFR_{dt}$ is the dismissal-induced increase in the student/faculty ratio.

The first-stage equation for student/faculty ratio is

$$SFR_{dt} = \delta_0 + \delta_1 DQ_{dt} + \delta_2 DSFR_{dt} + \Delta X_{idt} + \lambda_t + \mu_d + v_{idt}.$$

- $DismissalQualityFall_{dt}$ should predict lower average faculty quality after dismissals.
- $DismissalStudentFacultyIncrease_{dt}$ should predict a higher student/faculty ratio after faculty losses.
- The paper finds exactly this pattern: the two instruments predict the two endogenous variables in the expected way.

Interpretation: The first stage is the mechanical and historical foundation of the paper. The expulsions must actually move faculty quality and department size in order to identify student-outcome effects.

4.3 Reduced-form model

A reduced-form version regresses student outcomes directly on the dismissal-induced shocks:

$$Outcome_{idt} = \theta_0 + \theta_1 DQ_{dt} + \theta_2 DSFR_{dt} + \Theta X_{idt} + \lambda_t + \mu_d + e_{idt}.$$

- A negative coefficient on the dismissal-induced quality fall means students in departments that lost high-quality faculty had worse outcomes after the shock.

- The paper also performs a placebo test that moves the dismissal date to 1930 and uses only predissmissal observations.
- The placebo effects are close to zero and statistically insignificant, supporting the parallel-trends logic.

Interpretation: The reduced form shows the outcome consequences of the historical shock before translating them into the effect of measured faculty quality.

4.4 Instrumental variables interpretation

The IV estimator uses dismissal-induced changes to estimate the effect of faculty quality and student/faculty ratio on student outcomes.

- The IV effect is local to departments whose quality was affected by the dismissals.
- Because the dismissals hit many of the strongest departments, the IV estimates are especially informative about high-quality research environments.
- The paper finds that IV estimates are larger than OLS estimates, consistent with measurement error in observed faculty quality and with larger returns to quality in top departments.

Interpretation: The IV estimate should be read as the effect of exogenously losing or gaining high-quality faculty in the affected research departments, not as a universal effect for every possible department.

5 Identification and empirical design

5.1 What creates identification

- The Nazi dismissals created a sharp, externally imposed shock to faculty composition.
- Departments were affected very differently. Some lost high-quality professors, some lost lower-quality professors, and many lost none.
- Department fixed effects compare a department to itself before and after the shock.
- Cohort fixed effects compare treated departments to untreated departments within the same PhD cohort.

Interpretation: The design is a difference-in-differences style IV design. It asks whether student outcomes fell more in departments where the dismissals caused larger declines in faculty quality.

5.2 Main exclusion concern

- The key concern is that the dismissals may have affected students through channels other than faculty quality and student/faculty ratio.
- Possible channels include general disruption, loss of a specific dissertation adviser, differential treatment of Jewish students, or changes in department reputation.

- The paper addresses these concerns with sample restrictions and robustness checks.

Interpretation: The design is strongest if the relevant post-1933 differences across departments operate mainly through the removal of faculty quality rather than through other department-specific shocks.

5.3 Threats and responses

- **Jewish student selection.** Departments with Jewish professors may have attracted Jewish students, whose outcomes worsened under Nazi rule. The paper shows results are robust when Jewish students are excluded.
- **Disruption during dismissal years.** The paper omits 1933 and 1934 and finds results remain strong, often larger.
- **Specific-adviser loss.** The paper focuses on students still doing coursework in early 1933, before they had necessarily started the dissertation with a specific adviser. Results remain similar.
- **Differential trends.** The paper includes department-specific time trends and runs a placebo shock in 1930. The placebo does not predict outcomes.
- **Student switching.** The paper assigns post-1933 students to the department attended at the beginning of 1933 to avoid endogenous switching after the shock.

Interpretation: No single robustness check proves the design, but together they make it harder to attribute the results to simple pre-trends, Jewish-student composition, or temporary administrative disruption.

5.4 What the design identifies

- The strongest interpretation is that high-quality faculty improve PhD student human capital and career outcomes.
- The result is not simply that high-quality departments have better students; the paper uses within-department declines caused by expulsions.
- The results are most relevant for research-intensive departments and students who are strongly connected to research careers.
- The paper cannot fully rule out reputational channels, but it argues that abrupt one-year reputation changes are unlikely to explain the full magnitude of the results.

Interpretation: The paper provides evidence that faculty quality during PhD training has real productive value, not just signaling value.

6 Tables and figures

6.1 Figure 1 and Table 1: Motivation and timing of the shock

Read:

- Figure 1 shows the raw positive relationship between university quality and PhD student outcomes. In pre-1933 Germany, departments with higher faculty quality had higher probabilities that students published dissertations in top journals.
- The U.S. panel shows a similar positive cross-sectional relationship between department quality and placement rates in modern U.S. mathematics departments.
- Table 1 shows the timing of the dismissals. The key shock is concentrated in 1933–1934: 35 mathematics professors were dismissed in 1933 and 6 in 1934.
- By 1933–1934, 41 mathematics professors had been dismissed, equal to 18.3% of all mathematics professors in 1933.

Economic message: Figure 1 motivates the question, but is only correlational. Table 1 supplies the timing of the quasi-experimental shock that the paper later uses to move beyond correlation.

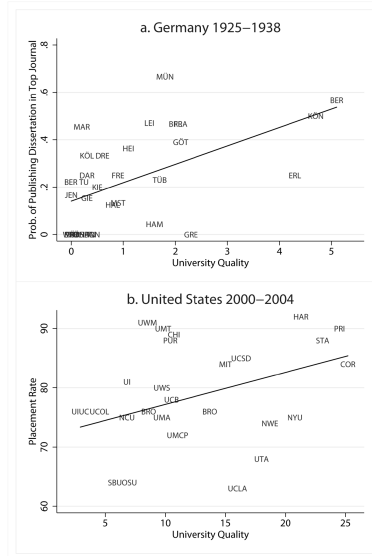


FIG. 1.—University quality and PhD student outcomes. *a.* Germany for the years 1925–38. The vertical axis reports the average probability of publishing the PhD dissertation in a top journal. The horizontal axis measures faculty quality as average department citations (see Sec. II D for details). *b.* United States for the years 2000–2004. The vertical axis reports the fraction of PhD students who have secured a permanent job or postdoc at the time of graduation. The horizontal axis measures faculty quality as average department citations (see <http://www.phds.org> and National Research Council [1995]).

TABLE 1
NUMBER OF DISMISSED MATHEMATICS PROFESSORS

Year of Dismissal	Number of Dismissed Professors	Percentage of All Mathematics Professors in 1933
1933	35	15.6
1934	6	2.7
1935	5	2.2
1936	1	.4
1937	2	.9
1938	1	.4
1939	1	.4
1940	1	.4
1933–34	41	18.3

Figure 1: University quality and PhD student outcomes

Table 1: Number of dismissed mathematics professors

6.2 Table 2 and Table 3: Cross-department variation and quality of dismissed faculty

Read:

- Table 2 shows large cross-department variation in the dismissal shock. Göttingen lost 10 of 17 professors, Berlin lost 5 of 13, and Breslau lost 3 of 6. Many departments lost no mathematics professors.
- The table also reports whether dismissals lowered, raised, or left unchanged average department quality.

- Table 3 shows that the dismissed professors were not a random subset of faculty. They represented 18.3% of professors but 31.1% of publications and 46.8% of citation-weighted publications.
- Dismissed professors were younger on average than stayers and much more productive by the citation-weighted measure.

Economic message: The shock was both heterogeneous and quality-relevant. The identification comes from departments losing different amounts of high-quality faculty.

TABLE 2
DISMISSALS ACROSS DIFFERENT UNIVERSITIES

UNIVERSITY	NUMBER OF PROFESSORS BEGINNING OF 1933	DISMISSED 1933-34		DISMISSAL- INDUCED CHANGE TO DEPARTMENT QUALITY
		Number	Percentage	
Aachen TU	7	3	42.9	+
Berlin	13	5	38.5	--
Berlin TU	14	2	14.3	+
Bonn	7	1	14.3	+
Braunschweig TU	3	0	0	0
Breslau	6	3	50.0	--
Breslau TU	5	2	40.0	--
Darmstadt TU	9	1	11.1	+
Dresden TU	10	0	0	0
Erlangen	3	0	0	0
Frankfurt	8	1	12.5	+
Freiburg	9	1	11.1	--
Gießen	7	1	14.3	+
Göttingen	17	10	58.8	--
Greifswald	3	0	0	0
Halle	7	1	14.3	+
Hamburg	8	0	0	0
Hannover TU	6	0	0	0
Heidelberg	5	1	20.0	+
Jena	5	0	0	0
Karlsruhe TU	6	1	16.7	0
Kiel	5	2	40.0	+
Köln	6	2	33.3	+
Königsberg	5	2	40.0	--
Leipzig	8	2	25.0	--
Marburg	8	0	0	0
München	9	0	0	0
München TU	5	0	0	0
Münster	5	0	0	0
Rostock	2	0	0	0
Stuttgart TU	6	0	0	0
Tübingen	6	0	0	0
Würzburg	4	0	0	0

NOTE.—This table reports the total number of professors in 1933. Number dismissed indicates how many professors were dismissed in each department. Percentage dismissed indicates the percentage of dismissed professors in each department. The column dismissal-induced change to department quality indicates how the dismissal affected average department quality: -- indicates a drop in average department quality by more than 50 percent, - a drop in average department quality between 0 and 50 percent, 0 no change in department quality, and + an improvement in average department quality between 0 and 50 percent.

TABLE 3
QUALITY OF DISMISSED PROFESSORS

	ALL	STAYERS	DISMISSED 1933-34	
			Number	Percentage Loss
Number of professors (beginning of 1933)	224	183	41	18.3
Number of chaired professors	117	99	18	15.4
Average age (1933)	48.7	50.0	43.0	. . .
Average publications (1925-32)	.33	.27	.56	31.1
Average citation-weighted publications (1925-32)	1.45	.93	3.71	46.8

NOTE.—Percentage loss is calculated as the fraction of dismissed professors among all professors or as the fraction of publications and citations that were contributed by the dismissed professors.

Table 2: Dismissals across different universities

Table 3: Quality of dismissed professors

6.3 Table 4 and Figure 2: Student data and first-stage visual evidence

Read:

- Table 4 summarizes the 690 PhD students. The average age at PhD is 27.5, about 8.7% are female, and 7.5% are foreign.
- About 24.1% publish the dissertation in a top journal, 18.7% later become chaired professors, and 29.9% have positive lifetime citations.
- Students from top 10 departments have better raw outcomes: 29.5% publish in top journals versus 18.3% in lower-ranked departments, and 25.0% become full professors versus 12.1%.
- Figure 2 shows the first-stage logic graphically. Departments with dismissals lose faculty size after 1933, and departments with above-average-quality dismissals experience a sharp fall in average faculty quality.

Economic message: The raw student outcome gradient is large but not causal. Figure 2 shows why the dismissal shock can be used as an instrument for quality and capacity.

TABLE 4
SUMMARY STATISTICS PhD STUDENTS

	All	Obtaining PhD in Top 10 Department	Obtaining PhD in Lower-Ranked Department
Average age at PhD ^a	27.5	26.9	28.1
Average time to PhD in years (from beginning of studies) ^b	7.4	7.2	7.7
% female	8.7	8.2	9.2
% foreign	7.5	9.7	5.3
Average number of university changes (from beginning of studies) ^c	1.1	1.1	1.2
Outcomes:			
% obtaining high school teacher diploma as additional degree	51.6	42.3	61.2
% published dissertation in top journal	24.1	29.5	18.3
% became chaired professor later in life	18.7	25.0	12.1
Average lifetime citations	11.2	16.6	5.5
% with positive lifetime citations	29.9	37.8	21.6
% with lifetime citations > 25	9.0	13.1	4.7
% with lifetime citations > 100	3.2	4.8	1.5
Number of PhD students	690	352	338

NOTE.—Summary statistics are based on all 690 PhD students in the sample.
^a Information on average age at PhD is available for 667 individuals.
^b Average time to PhD is available for 579 individuals.
^c Information on the number of university changes is available for 626 individuals.

Table 4: Summary statistics for PhD students

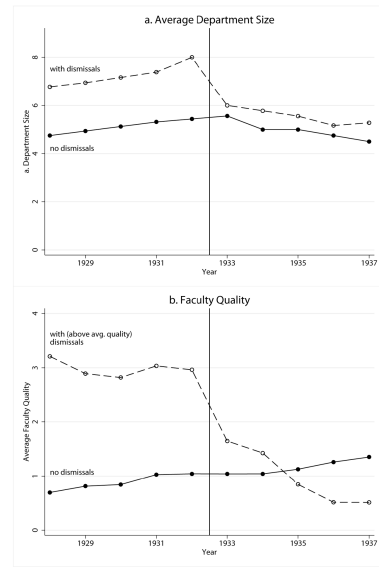


FIG. 2.—Effect of dismissals on faculty in mathematics departments. a. Effect on department size. Dashed line: departments with dismissals in 1933 and 1934; solid line: departments without dismissals. b. Effect on average faculty quality. Dashed line: departments with dismissals of above-average department quality between 1933 and 1934; solid line: departments without dismissals.

Figure 2: Effect of dismissals on faculty in mathematics departments

6.4 Figure 3 and Table 5: Reduced-form outcomes and first stages

Read:

- Figure 3 plots outcome trends for students in affected versus unaffected departments. After 1933, students in departments with above-average-quality dismissals show worse outcomes, especially for dissertation publication and lifetime citations.
- Table 5 reports the first-stage regressions. The dismissal-induced fall in faculty quality strongly predicts lower average faculty quality.
- The dismissal-induced increase in student/faculty ratio strongly predicts the student/faculty ratio.
- The Cragg-Donald statistic is 25.2, above the relevant weak-instrument critical value discussed in the paper.

Economic message: Figure 3 shows the reduced-form damage to student outcomes. Table 5 shows that the instruments have the expected mechanical relationship with the endogenous regressors.

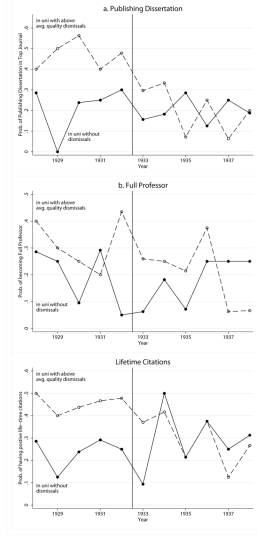


FIG. 3.—Effect of dismissals on PhD student outcomes. *a*, Effect on the probability of publishing dissertation in a top journal. Dashed line: departments with above-average quality dismissals between 1933 and 1934; solid line: departments without dismissals. *b*, Effect on probability of becoming a full professor later in life. Dashed line: departments with above-average quality dismissals between 1933 and 1934; solid line: departments without dismissals. *c*, Effect on the probability of having positive lifetime citations. Dashed line: departments with above-average quality dismissals between 1933 and 1934; solid line: departments without dismissals.

Figure 3: Effect of dismissals on PhD student outcomes

TABLE 5
FIRST STAGES

	DEPENDENT VARIABLE	
	Average Quality (1)	Student/Faculty Ratio (2)
Dismissal-induced fall in faculty quality	-1.236** (.074)	-4.195 (2.058)
Dismissal-induced increase in student/faculty ratio	.014 (.008)	.439** (.116)
Female	.142* (.060)	1.165 (.705)
Foreigner	.046 (.097)	-1.971 (1.183)
Cohort dummies	Yes	Yes
University fixed effects	Yes	Yes
Observations	690	690
R^2	.795	.757
Cragg-Donald eigenvalue statistic	25.2	

NOTE.—All standard errors are clustered at the department level.

*Significant at the 5 percent level.

**Significant at the 1 percent level.

Table 5: First stages

6.5 Table 6 and Table 7, part 1: Main IV estimates and robustness

Read:

- Table 6 is the core IV result. Average faculty quality has positive and statistically significant effects on all four student outcomes.
- The IV coefficient for publishing in a top journal is 0.102, implying that a one-standard-deviation increase in faculty quality raises the probability by about 13 percentage points.
- The IV coefficient for becoming a full professor is 0.076, implying an increase of about 10 percentage points for a one-standard-deviation quality increase.
- The IV coefficient for lifetime citations is 4.901, implying an increase of about 6.3 citations for a one-standard-deviation quality increase.
- Student/faculty ratio is not a robust predictor of student outcomes.
- Table 7 begins the robustness analysis. The estimates remain similar when adding father's occupation, focusing on earlier-starting students, excluding Jewish students, omitting 1933–1934, and focusing on students still doing coursework.

Economic message: The main estimated effect is large and is driven by faculty quality, not by simple faculty/student ratios. The first part of the robustness table shows that the result is not fragile to major sample changes.

	INSTRUMENTAL VARIABLES							
	DEPENDENT VARIABLE							
	Published Top		Full Professor		Number of Lifetime Citations		Positive Lifetime Citations	
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)
Average faculty quality	.056** (.018)	.102** (.015)	.037 (.021)	.076** (.015)	2.388* (1.119)	4.901** (1.546)	.068* (.026)	.125** (.016)
Student/faculty ratio	.000 (.001)	.003 (.002)	.000 (.001)	-.001 (.003)	.040 (.106)	-.284 (.328)	.002 (.001)	.003 (.004)
Female	-.015 (.059)	-.022 (.055)	-.099* (.041)	-.103** (.036)	-8.711** (2.736)	-8.850** (2.436)	-.090 (.068)	-.099 (.062)
Foreigner	.014 (.048)	.022 (.045)	-.134* (.053)	-.135* (.053)	.803 (5.136)	.016 (5.262)	-.025 (.072)	-.019 (.074)
Cohort dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	690	690	690	690	690	690	690	690
R ²	.163		.155		.080		.126	
Cragg-Donald eigenvalue statistic		25.2		25.2		25.2		25.2

NOTE.—All standard errors are clustered at the department level.
*Significant at the 5 percent level.
**Significant at the 1 percent level.

Table 6: Instrumental variables results

	TABLE 7 ROBUSTNESS INSTRUMENTAL VARIABLE RESULTS						
	SAMPLE						
	Full Sample	Full Sample	Starting 1922–30	Non-Jewish Students	Omitting 1933 and 1934	Students Doing Course Work	Full Sample
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Published top:							
Average faculty quality	.102** (.013)	.109** (.013)	.096** (.019)	.094** (.018)	.124** (.019)	.144** (.040)	.048* (.022)
Student/faculty ratio	.003 (.002)	.001 (.003)	.002 (.003)	.002 (.003)	.001 (.003)	.008 (.014)	.004 (.005)
Full professor:							
Average faculty quality	.076** (.013)	.082** (.016)	.057** (.014)	.065** (.019)	.091** (.020)	.069* (.033)	.055** (.019)
Student/faculty ratio	-.001 (.003)	-.002 (.004)	-.002 (.005)	-.004 (.003)	-.004 (.003)	.013 (.008)	-.006 (.003)
Number of lifetime citations:							
Average faculty quality	4.901** (1.540)	6.038** (1.658)	5.588** (1.684)	4.966** (1.233)	6.102** (2.078)	5.448** (1.973)	4.540 (2.922)

Table 7: Robustness IV results, part 1

6.6 Table 7, part 2 and Table 8: Robustness continued and heterogeneity

Read:

- The continuation of Table 7 reports the lower half of the robustness table, including results for positive lifetime citations, controls, sample sizes, and Cragg-Donald statistics.
- Department-specific time trends reduce some estimates, especially the lifetime-citation outcomes, but the dissertation publication and full-professor results remain positive.
- Table 8 splits the sample between top 10 departments and lower-ranked departments.
- Returns to faculty quality are concentrated in top departments. The instruments are strong in top departments but weak in lower-ranked departments.

Economic message: The effect of faculty quality is strongest where research training is most intensive. This supports the interpretation that high-quality research environments generate large returns for PhD students.

Student/faculty ratio	-.284 (.328)	-.272 (.247)	-.272 (.353)	.213 (.291)	-.334 (.375)	.246 (.446)	-.058 (.296)
Positive lifetime citations:							
Average faculty quality	.125** (.016)	.136** (.017)	.129** (.018)	.106** (.020)	.130** (.028)	.174** (.040)	.012 (.030)
Student/faculty ratio	.003 (.004)	.000 (.004)	-.003 (.003)	.001 (.004)	.001 (.004)	.004 (.013)	-.010 (.011)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Father's occupation	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department-specific time trends							
Observations	690	690	542	635	570	501	690
Cragg-Donald eigenvalue statistic	25.2	21.6	18.7	23.5	17.1	6.4	15.1

NOTE.—This table shows only regressors of interest. Results in the four panels correspond to separate regressions with an indicator for publishing the dissertation in a top journal, an indicator for becoming a full professor, number of lifetime citations, and an indicator for positive lifetime citations as the respective dependent variables. Regressors not listed are indicators for being female and foreigner. See the text for details. All standard errors are clustered at the department level.
*Significant at the 5 percent level.
**Significant at the 1 percent level.

Table 7: Robustness IV results, part 2

DEPENDENT VARIABLE	SAMPLE			
	Top 10 Department		Lower-Ranked Department	
	OLS (1)	IV (2)	OLS (3)	IV (4)
Published top:				
Average faculty quality	.059** (.014)	.094** (.022)	.012 (.032)	-.214 (.259)
Student/faculty ratio	-.001 (.002)	.002 (.003)	.001 (.002)	.016 (.026)
Full professor:				
Average faculty quality	.046* (.017)	.074** (.019)	-.086* (.034)	-.065 (.214)
Student/faculty ratio	-.001 (.002)	-.001 (.003)	.001 (.003)	-.000 (.023)
Number of lifetime citations:				
Average faculty quality	2.092* (.685)	2.667** (.557)	-4.570 (3.396)	-6.032 (24.717)
Student/faculty ratio	.071 (.167)	.156 (.211)	.023 (.156)	.158 (2.914)
Positive lifetime citations:				
Average faculty quality	.092** (.016)	.138** (.020)	-.055 (.043)	-.164 (.204)
Student/faculty ratio	-.001 (.002)	.003 (.003)	.003 (.002)	.007 (.020)
Controls	Yes	Yes	Yes	Yes
Cohort dummies	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes
Observations	352	352	338	338
Cragg-Donald eigenvalue statistic		30.0		.8

NOTE.—This table shows only regressors of interest. Results in the four panels correspond to separate regressions with an indicator for publishing the dissertation in a top journal, an indicator for becoming a full professor, number of lifetime citations, and an indicator for positive lifetime citations as the respective dependent variables. Regressors not listed are indicators for being female and foreigner. See the text for details. All standard errors are clustered at the department level.
*Significant at the 5 percent level.
**Significant at the 1 percent level.

Table 8: Heterogeneity in returns

6.7 Appendix Table A1 and Appendix Table A2: Measurement details

Read:

- Table A1 lists the top journals used to construct publication and citation measures. The list includes German mathematics journals, international mathematics journals, and related journals in general science, physics, and chemistry.
- Table A2 lists the top 20 mathematics professors by citation-weighted publications between 1925 and 1932.
- The highest-ranked names include John von Neumann, Richard Courant, Richard von Mises, and other leading mathematicians.
- Many of the most cited professors were dismissed in 1933–1934.

Economic message: The appendix tables show that the quality measure is anchored in recognizable scholarly output and that the dismissals removed a disproportionate share of elite mathematicians.

Appendix

TABLE A1
TOP JOURNALS

Journal Name	Published in
Mathematics:	
Journal für die reine und angewandte Mathematik	Germany
Mathematische Annalen	Germany
Mathematische Zeitschrift	Germany
Zeitschrift für angewandte Mathematik und Mechanik	Germany
Acta Mathematica	Sweden
Journal of the London Mathematical Society	United Kingdom
Proceedings of the London Mathematical Society	United Kingdom
General science:	
Naturwissenschaften	Germany
Sitzungsberichte der Preussischen Akademie der Wissenschaften	
Physikalisch Mathematische Klasse	Germany
Nature	United Kingdom
Proceedings of the Royal Society of London A (Mathematics and Physics)	United Kingdom
Science	United States
Physics:	
Annalen der Physik	Germany
Physikalische Zeitschrift	Germany
Physical Review	United States
Chemistry:	
Berichte der Deutschen Chemischen Gesellschaft	Germany
Biochemische Zeitschrift	Germany
Journal für Praktische Chemie	Germany
Justus Liebig's Annalen der Chemie	Germany
Kolloid Zeitschrift	Germany
Zeitschrift für Anorganische Chemie und Allgemeine Chemie	Germany
Zeitschrift für Elektrochemie und Angewandte Physikalische Chemie	Germany
Zeitschrift für Physikalische Chemie	Germany
Journal of the Chemical Society	United Kingdom

Table A1: Top journals used for the quality measure

TABLE A2
TOP 20 MATHEMATICS PROFESSORS, 1925–32: CITATION-WEIGHTED PUBLICATIONS MEASURE

Name	University Beginning of 1933	Average Citation-Weighted Publications (1925–32)	Average Publications (1925–32)	Dismissed 1933–34
John von Neumann	Berlin	36.3	1.5	Yes
Richard Courant	Göttingen	22.3	1.3	Yes
Richard von Mises	Berlin	15.6	.9	Yes
Heinz Hopf ^a		13.3	1.3	
Paul Epstein	Frankfurt	11.5	.6	
Oskar Perron	München	10.6	1.5	
Willy Prager	Göttingen	10.0	.4	Yes
Gábor Szegő	Königsberg	9.4	1.4	Yes
Werner Rogosinski	Königsberg	9.1	.6	
Wolfgang Krull	Erlangen	8.9	1.4	
Erich Rothe	Breslau TU	8.0	1.0	Yes
Hans Petersson	Hamburg	8.0	2.0	
Adolf Hammerstein	Berlin	8.0	.5	
Alexander Weinstein	Breslau TU	6.3	.7	Yes
Erich Kamke	Tübingen	6.3	.8	
Hellmuth Kneser	Greifswald	6.3	.6	
Bartel van der Waerden	Leipzig	5.8	1.8	
Max Müller	Heidelberg	5.3	.3	
Richard Brauer	Königsberg	5.0	.6	Yes
Leon Lichtenstein	Leipzig	4.9	1.5	Yes

^a The university in 1933 is missing for professors who retired before 1933.

Table A2: Top 20 mathematics professors by citation-weighted output

6.8 Appendix Table A3 and Appendix Figure A1: Reduced form, placebo, and source material

Read:

- Table A3 reports reduced-form estimates and the placebo test moving the dismissal shock to 1930.
- The true reduced form shows that dismissal-induced quality losses significantly reduce student outcomes.
- The placebo estimates using only predissmissal observations are not statistically significant, which supports the claim that treated and untreated departments were not already diverging before 1933.
- Figure A1 shows a sample page from the *List of Displaced German Scholars*, one of the main sources for identifying dismissed mathematicians.

Economic message: The reduced-form and placebo evidence are central to the identification argument. The historical source image shows how the paper turns archival records into quantitative treatment variation.

	REDUCED-FORM AND PLACEBO TEST							
	DEPENDENT VARIABLE							
	Reduced Form				Placebo Moving Dismissal to 1930 (Only Pre-1933 Observations)			
	Published Top (1)	Full Professor (2)	No. of Lifetime Citations (3)	Positive Lifetime Citations (4)	Published Top (5)	Full Professor (6)	No. of Lifetime Citations (7)	Positive Lifetime Citations (8)
Dismissal-induced fall in faculty quality	-.134** (.017)	-.090** (.021)	-6.157** (2.218)	-.164** (.019)	-.023 (.031)	.053 (.037)	3.454 (3.597)	-.037 (.030)
Dismissal-induced increase in student/faculty ratio	.002 (.001)	.000 (.001)	-.042 (.114)	.002 (.002)	-.004 (.004)	-.002 (.005)	-.462 (.451)	-.002 (.003)
Female	.004 (.048)	-.119* (.045)	-10.723* (4.459)	-.067 (.058)	.009 (.066)	-.167* (.068)	-12.114** (4.228)	-.104 (.071)
Foreigner	.031 (.048)	-.147* (.065)	.942 (6.151)	-.033 (.075)	-.017 (.103)	-.136 (.102)	-7.169 (6.479)	-.050 (.134)
Father's occupation	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	690	690	690	690	403	403	403	403
R ²	.221	.208		.185	.208	.202	.291	.224

NOTE.—All standard errors are clustered at the department level.
*Significant at the 5 percent level.
**Significant at the 1 percent level.

Table A3: Reduced form and placebo test

Mathematics	
M A T H E M A T I C S	
BAER, Dr. Reinhold, Privatdozent; b. 02., married, 4 children. (English, French, Italian.) 1909/30: Assistant, later Privatdozent Freiburg University; 1929/33: Privatdozent Halle University; 1934/34: "Researcher" Manchester University; since 1935: Institute for Advanced Study, Princeton (N.J.). SPEC.: <i>Pure Math.; Theory of Functions; Algebraic Theory of Topology; Algebraic Theory of Differential Equations</i> . Temp.	BREUER, Dr. Samuel, a.o. Professor; b. 01., married, 8 children. (English, French, Italian.) 1921/33: Privatdozent, later a.o. Prof. Technische Universität Karlsruhe; 1929/33: Assistant Frankfurt University; since 1934: Industrial Activity, Jerusalem. SPEC.: <i>Pure and Applied Math.; Algebra; Insurance; Statistics</i> . Perm.
BAERWALD, Hans Georg, see ENGINEERING. PHYSICS.	COUBANT, Dr. Richard, a. Professor; b. 08., married, 4 children. (English.) 1912/30: Privatdozent, later o. Prof. Göttingen University; 1933/34: Lecturer Cambridge University; since 1938: New York University. SPEC.: <i>Pure Math.</i> Perm.
BERGMANN, Dr. Stephan, Privatdozent; b. 07., single. 1922/33: Assistant Institut für Angewandte Mathematik, Berlin University; 1934/37: Researcher and Industrial Activity, Lodz; 1937/38: Privatdozent Berlin University; since 1934: State University, Toronto. SPEC.: <i>Pure and Applied Math.; Theory of Elasticity; Conformal Representation; Differential Equations; Hydrodynamics; Theory of Functions</i> . Perm.	DEHN, Dr. Max, a. Professor; b. 78., married, 1 child. (English, French, Norwegian.) 1911/13: a.o. Prof. Kiel University; 1913/21: a. Prof. Technische Hochschule, Bismar; 1921/35: a. Prof. Frankfurt University. SPEC.: <i>Pure Math.; Analysis; Algebra; History of Mathematics</i> . Unpl.
BERNAYS, Dr. Paul, a.o. Professor; b. 08., single. (English, French.) 1917/33: Privatdozent, later a.o. Prof. Göttingen University and Assistant Mathematisch-Physikalisches Seminar; 1934/38: Lecturer Technische Hochschule, Zürich; 1938/39: Visiting Lecturer Institute for Advanced Study, Princeton (N.J.); 1938/37: Eidgenössisches Polytechnikum, Zürich. SPEC.: <i>Logic; Foundation of Mathematics</i> . Temp.	DINGLER, Dr. Hugo, see PHILOSOPHY.
BERNSTEIN, Dr. Felix, a. Professor; b. 78., married, 2 children. (English.) 1904/08: Privatdozent Halle University; 1908/33: Privatdozent, later o. Prof. Göttingen University; since 1934: Columbia University, New York. SPEC.: <i>Statistics; Group Theory; Calculation of Probability; Insurance; Finance</i> . Temp.	FELLER, Dr. Willy, Privatdozent; b. 06., single. (English, French, Croatian.) 1906/29: Assistant Göttingen University; since 1934: Assistant Kiel University; since 1934: Institute of Mathematics, Stockholm University. SPEC.: <i>Pure and Applied Math.; Differential Geometry; Modern Theory of Real Functions</i> . Temp.
BILMUTHAL, Dr. Otto, a. Professor; b. 76., married, 2 children. (English, French, Italian, Russian, Dutch, Hungarian.) 1901/04: Privatdozent Göttingen University; 1908/31: a. Prof. Technische Hochschule, Aachen; since 1909: Editor "Mathematische Annalen." SPEC.: <i>Pure and Applied Math.; Theory of Functions; Hydrodynamics</i> . Unpl.	FENCHEL, Dr. Werner, Assistant; b. 05., married. (English, Italian, Danish.) Till 1933: Assistant Göttingen University; since 1934: Mathematisk Institut, Copenhagen. SPEC.: <i>Pure and Applied Math.; Geometry; Analysis; Mathematical Methods of Economics</i> . Temp.
BOCHNER, Dr. Salomon, Dozent; b. 09. Till 1933: Dozent, Munich University; since 1933: Princeton University. Perm.	FRAENKEL, Dr. Adolf, a. Professor; b. 01., married, 4 children. 1918/28: Privatdozent, later a.o. Prof. Marburg University; 1928/33: a. Prof. Kiel University; since 1933: Prof. Hebrew University, Jerusalem. SPEC.: <i>Pure Math.; Theory of Sets; Arithmetic; Borderlines of Philosophy</i> . Perm.
BOCHNER, Dr. Richard, Privatdozent; b. 01., married, 2 children. (English, French.) 1925/33: Assistant, later Privatdozent Königslberg University; 1934/38: Assistant Institute for Advanced Study, Princeton (N.J.); since 1938: Assistant Prof. Toronto University. SPEC.: <i>Pure Math.; Algebra; Group Theory; Theory of Hypersurfaces; Numbers</i> . Perm.	GOLOMB, Dr. Michael, Researcher; b. 09., single. (English, French, Italian, Spanish.) 1929: Researcher Berlin University; since 1934: Zagreb. SPEC.: <i>Pure Math.; Analysis of Functions; Differential and Integral Equations</i> . Unpl.
	GRADSTEIN, Dr. Stephan, see PHYSICS. ENGINEERING.
	GUMBEL, Dr. Emil J., a.o. Professor; b. 01., married, 1 child. (English, French, Italian.) 1928/33: Privatdozent, later a.o. Prof. Heidelberg University; since 1937: Lyons University. SPEC.: <i>Applied Math.; Statistics; Calculus of Probability; Actuarial Science</i> . Perm.

FIG. A1.—Sample page from the mathematics section of the *List of Displaced German Scholars* (1936).

6.9 Appendix Figure A2: Jewish PhD students

Read:

- Figure A2 reports the number of Jewish mathematics PhD graduates by year.
- The number is small in the years before 1933 and declines after the Nazi regime's policies take effect.
- The paper uses this evidence to motivate the robustness check excluding Jewish students.

Economic message: Jewish student selection is a plausible concern, but the figure shows that the number of Jewish PhD graduates is small. The non-Jewish-only robustness results further reduce this concern.

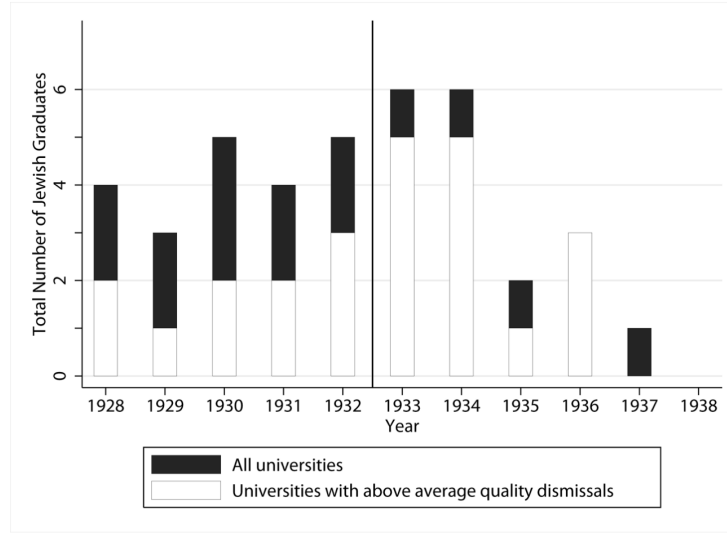


FIG. A2.—Total number of Jewish mathematics PhD graduates in all German universities in each year. Black bar: all departments; white bar: departments with above-average quality dismissals between 1933 and 1934.

Figure A2: Total number of Jewish mathematics PhD graduates

7 Short summary

- Waldinger studies whether faculty quality causally affects PhD student outcomes.
- The empirical setting is mathematics in Germany before World War II.
- The key shock is the Nazi expulsion of Jewish and politically unreliable professors after 1933.
- The dismissals were severe, concentrated in 1933–1934, and hit departments unevenly.
- The dismissed professors were disproportionately high quality by citation-weighted publications.
- The paper constructs a student-level data set of mathematics PhDs and links it to department-year measures of faculty quality.
- The main specification includes department and cohort fixed effects and instruments faculty quality and student/faculty ratio with dismissal-induced changes.
- Faculty quality has large positive effects on dissertation publication, full-professor status, lifetime citations, and positive lifetime citations.
- Student/faculty ratio has little explanatory power once faculty quality is accounted for.
- Robustness checks show that the results are not driven by Jewish students, disruption in the dismissal years, advisor loss, or simple predissmissal trends.
- The effects are strongest in top departments, consistent with high returns to quality in research-intensive environments.

8 Conclusion

8.1 Core conclusion

Waldinger shows that faculty quality during PhD training has large effects on student outcomes. The paper’s central empirical finding is that losing high-quality professors reduced the research output and long-run academic success of PhD students. The estimated magnitudes are economically large: one standard deviation in faculty quality changes dissertation publication, full-professor status, and lifetime citations by amounts that are meaningful relative to the baseline outcomes.

8.2 Economic intuition

The economic intuition is that PhD training is a direct human-capital production process. High-quality professors provide better dissertation guidance, expose students to stronger research problems, transmit tacit knowledge, and connect students to scientific standards and networks. These inputs matter especially for young researchers, who have not yet developed their own professional networks. Senior professors may be less dependent on local department quality, but PhD students are highly exposed to the local intellectual environment.

8.3 Incremental contribution revisited

The paper advances the education and science-economics literatures in three main ways. First, it uses an externally imposed historical shock rather than cross-sectional program rankings. Second, it constructs a detailed measure of department quality based on faculty publication and citation records. Third, it follows students into long-run academic outcomes. This combination allows the paper to move from “better departments have better students” to “department faculty quality appears to causally shape research careers.”

8.4 Implications for PhD training and science policy

The findings imply that the organization of doctoral education matters. If high-quality faculty generate large benefits for PhD students, then spreading doctoral students thinly across many lower-quality departments may reduce scientific productivity. Concentrating PhD training in departments with strong research faculty can raise student output and long-run careers. This is why the paper discusses Göttingen and Berlin before World War II, and contrasts historical German concentration with the more dispersed postwar German system.

8.5 Limitations

The setting is historically unusual, and the shock occurred under extreme political conditions. External validity to modern universities is therefore not automatic. The paper observes completed PhDs, so it cannot fully measure effects on entry, dropout, or students who left the system. The faculty-quality measure is based on publications and citations, which are imperfect proxies for teaching, mentoring, and intellectual influence. Finally, reputational channels cannot be ruled out

completely, although the paper argues that abrupt reputation changes are unlikely to explain the full results.

8.6 Takeaway

The enduring takeaway is that quality matters in PhD training. Faculty quality is not only a prestige signal attached to a department name; it appears to be an input that changes students' human capital and long-run scientific output. The paper turns a tragic historical shock into evidence on how research environments shape the next generation of scientists.